

SOWMIYA M

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Summary

Computer Science undergraduate (2026 graduate) with strong fundamentals in object-oriented design, data structures, and algorithms, backed by 100+ solved DSA problems on LeetCode. Experienced in designing REST APIs, relational database schemas, and containerized backend systems using Java and Spring Boot. Skilled at building scalable, automated pipelines and comfortable working through broadly-defined, ambiguous problems end to end.

Education

Sri Krishna College of Technology
Bachelor of Science in Computer Science

Aug. 2024 – Sep. 2026
Coimbatore, Tamilnadu

Technical Skills

Core CS: Data Structures & Algorithms, Object-Oriented Design, Complexity Analysis, Problem Solving
Languages: Java, Python, C, C++, JavaScript, SQL
Backend & APIs: Spring Boot, REST API, Relational Databases (schema design, query optimization)
Systems & Automation: Distributed workflow automation (n8n), Selenium, API Integration
DevOps & Tools: Docker, Git, GitHub, VS Code, Linux
Cloud: Google Cloud Platform

Projects

Healthcare Appointment Mgmt System | *Spring Boot 3, Spring Security (JWT), JPA/Hibernate, MySQL* **2025**

- Designed and built a role-based backend (Patient, Doctor, Admin) with a normalized relational schema across User, Doctor, Patient, Appointment, and Schedule entities, mapped via Hibernate/JPA.
- Implemented stateless authentication and authorization using Spring Security 6 with JWT, enforcing endpoint-level access control by role.
- Modeled the appointment lifecycle as a status workflow (REQUESTED → APPROVED / REJECTED → COMPLETED) enforced through REST APIs and service-layer validation.
- Documented and tested all REST endpoints with Swagger/OpenAPI, and structured the project with Maven for build and dependency management.

Employee Issuance Check System | *Java, Spring Boot, REST API, MySQL, Docker* **2024**

- Built a layered Spring Boot backend exposing REST APIs to create, update, and query employee issuance records, backed by a relational database with indexed lookups by employee and status.
- Modeled issuance state (active / pending / completed) as an explicit status field with service-layer transition rules, and added automated notifications for pending/incomplete records.
- Containerized the application with Docker to standardize builds and simplify deployment across environments.

RAG-Based AI Agent System | *n8n, RAG, LLM, Vector Database* **2025**

- Designed a Retrieval-Augmented Generation pipeline combining document embedding, vector search, and LLM reasoning to answer knowledge-based queries.
- Built automated ingestion pipelines covering chunking, embedding generation, and semantic search over a vector database.

AI Agent for Payment Due Automation | *n8n, Google Sheets API, WhatsApp API* **2025**

- Built an event-driven workflow that tracks payment due dates from Google Sheets and triggers automated WhatsApp reminders, removing manual follow-up.

Achievements & Certifications

Smart India Hackathon (SIH) 2025: Qualified through four levels at the intercollege level.
MSME Hackathon 2025: Recognized at the intercollege level for innovative project development.
LeetCode: Solved 100+ Data Structures and Algorithms problems across arrays, trees, graphs, and dynamic programming.
Oracle Cloud Infrastructure: 2025 Certified AI Foundations Associate.
Microsoft Azure Fundamentals: Coursera.